

Real-Screen Pager Flash Elimination

THE PREVIEWED SCREEN IS THE COMMITTED SCREEN

Why the first revision still flashed

The stationary bottom-navigation revision removed the outgoing-page reset flash, but it still displayed a lightweight `SwipeTabPreview` during the drag. Releasing the gesture replaced that placeholder with the real destination component. The abrupt placeholder-to-screen substitution was the remaining visible flash reported by the user.

Before and after

Area	Previous revision	Corrected revision
Drag preview	Synthetic title, icon, text, and skeleton card.	The actual Home, Explore, Downloads, Nex AI, or Profile component.
Release handoff	Placeholder unmounted; real destination mounted or became visible.	The keyed destination instance remains mounted through commit.
Page retention	Only current content plus a disposable preview.	Visited or previewed root tabs remain in stable keyed layers.
Layer settling	Native-driver transforms could retain stale offscreen positions.	Pager transforms settle atomically on the JS driver already used by <code>PanResponder</code> .
Bottom navigation	Fixed, with color-only selected state.	Still fixed; only the active icon and label color changes.

Implementation

- Replaced children plus `SwipeTabPreview` with `renderTab(tab)`, which mounts the actual adjacent root screen.
- Retained mounted pages in stable `ROOT_TAB_ORDER` keys so destination components are not reparented or remounted at commit.
- Calculated idle and transition offsets for every retained page while keeping non-active pages non-interactive and accessibility-hidden.
- Moved pager timing and spring completion to the JavaScript driver to avoid stale native transform state during layer-role changes.
- Prevented Nex AI setup modals from opening while the AI tab is mounted only as an inactive preview.
- Removed the synthetic preview component and its unused styles.

Verification matrix

Check	Evidence	Result
All-frame review	Every decoded transition frame used real Home or Explore content; no placeholder, blank, or prior-page insertion appeared.	PASS

Check	Evidence	Result
Forward swipe	Home to Explore revealed Search courses here after a center-screen drag.	PASS
Reverse swipe	Explore to Home revealed Continue Learning without remount flash.	PASS
Short cancel	A sub-threshold drag returned to the same mounted Home page.	PASS
Carousel isolation	Recommended Lessons horizontal drag remained within the carousel.	PASS
Multi-tab route	Profile to Nex AI to Downloads passed with actual destination screens.	PASS
Bottom bar	One stationary bar remained; only selected color changed.	PASS
iOS export	Expo exported 731 modules and a 2.3 MB Hermes bundle.	PASS
Diff hygiene	git diff --check completed without whitespace errors.	PASS

Release assessment

The remaining placeholder-to-real-screen flash is removed. The adjacent destination visible while dragging is the same mounted component retained after release. No package, native dependency, app version, or iOS build-number change was required.

Known unchanged warning

React Native DevTools still reports the existing SafeAreaView deprecation warning. It is unrelated to the pager transition.